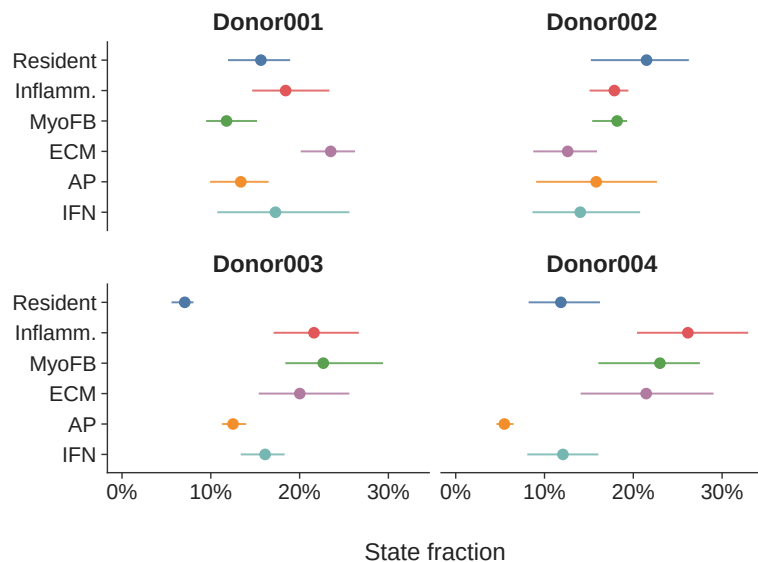
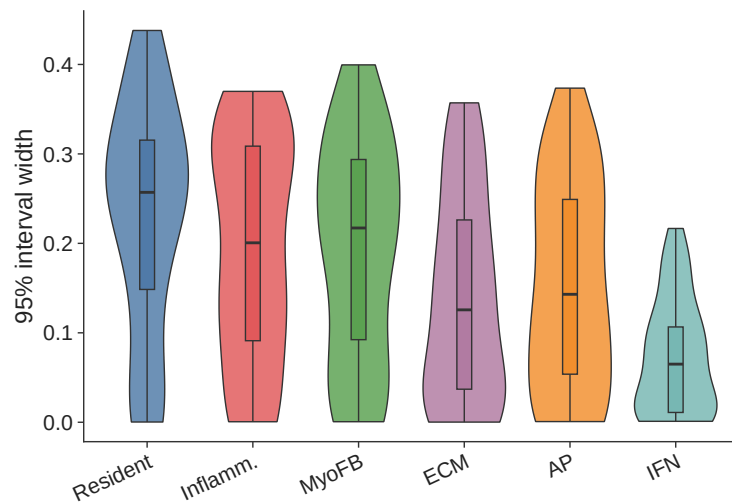


A**Bootstrap composition intervals**

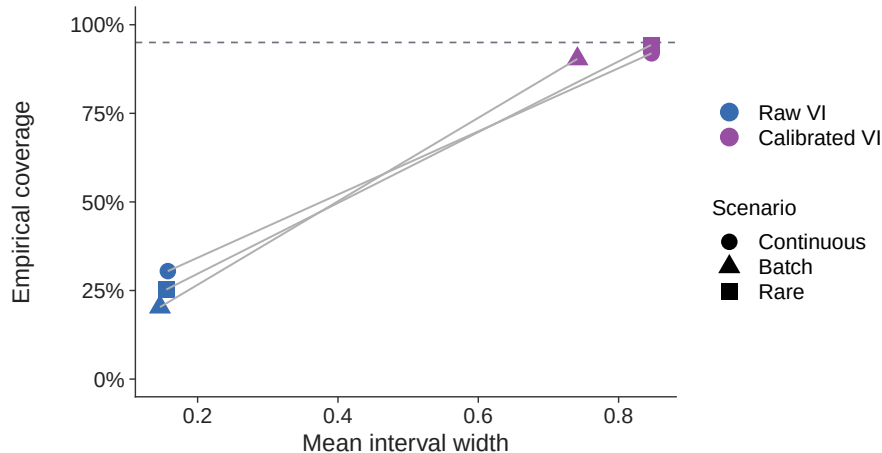
Sample-level state composition, mean and 95% interval

**B****VI state-weight interval width**

Cell-level logistic-normal posterior intervals

**C****Interval coverage calibration**

Dashed line marks nominal 95% coverage

**D****Uncertainty tracks mixed-state entropy**

Cells with higher entropy have wider intervals

